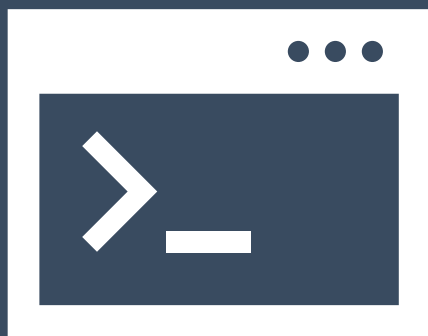


Introduction to Single-cell RNA-seq analysis

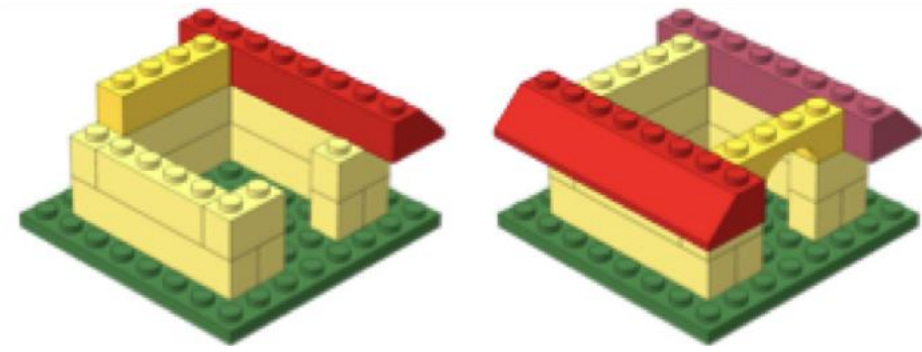
<https://tinyurl.com/hbc-intro-scrna>



Harvard Chan Bioinformatics Core



Workshop Scope



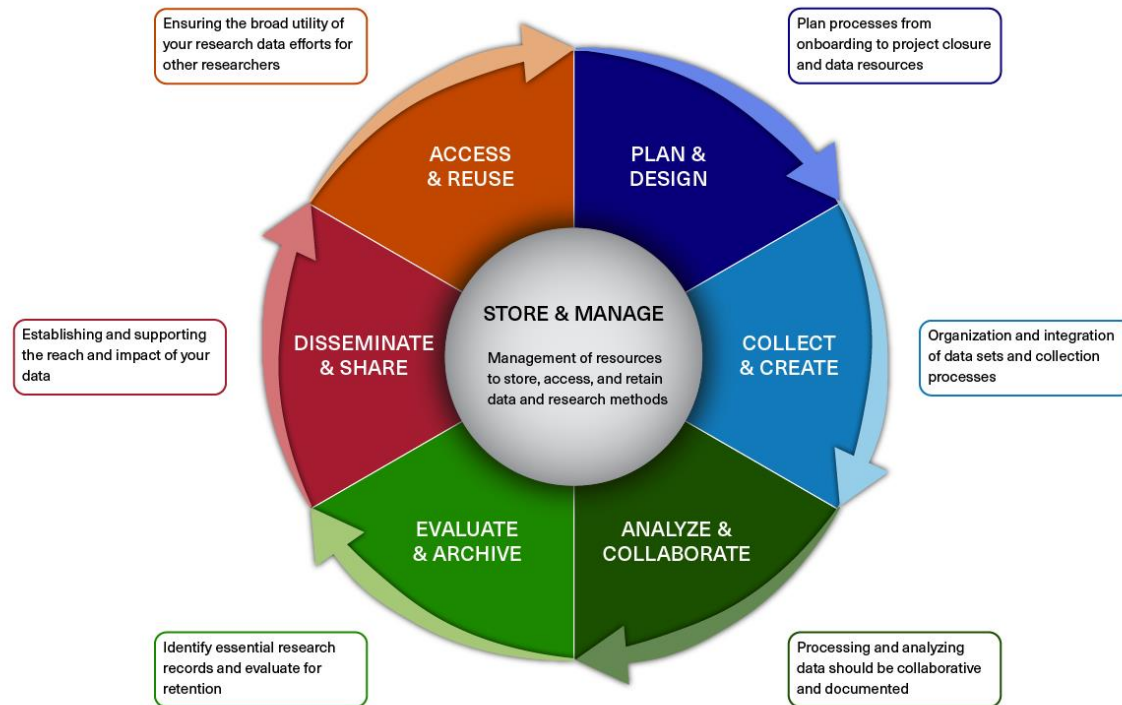
- ❖ Describe best practices for designing a single-cell RNA-seq experiment
- ❖ Describe steps in a single-cell RNA-seq analysis workflow
- ❖ Use Seurat and associated tools to perform analysis of single-cell expression data, including data filtering, QC, integration, clustering, and marker identification
- ❖ Understand practical considerations for performing scRNA-seq, rather than in-depth exploration of algorithm theory

Exit survey

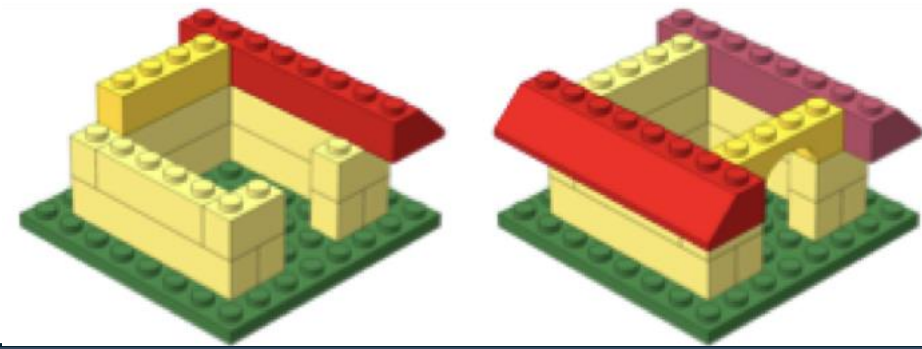
<https://tinyurl.com/scRNAseq-online>

Research Data Management (RDM)

BIOMEDICAL RESEARCH DATA LIFECYCLE



Useful resources



❖ Computational packages for single-cell analysis:

- ❖ <http://bioconductor.org/packages/devel/workflows/html/simpleSingleCell.html>
- ❖ <https://satijalab.org/Seurat/>
- ❖ <https://scanpy.readthedocs.io/>
- ❖ <https://github.com/seandavi/awesome-single-cell>

❖ Online courses:

- ❖ <https://hemberg-lab.github.io/scRNA.seq.course/>
- ❖ <https://github.com/SingleCellTranscriptomics>

❖ Resources for scRNA-seq Sample Prep:

- ❖ <https://www.protocols.io/>
- ❖ <https://support.10xgenomics.com/single-cell-gene-expression/sample-prep>
- ❖ <https://community.10xgenomics.com/>

Better RDM practice benefits you

❖ HMS Data Management LMA

❖ **Webpage:** <https://datamanagement.hms.harvard.edu>

❖ **Sign up** for quarterly email updates

❖ Harvard-wide Research data Management

❖ <https://researchdatamanagement.harvard.edu/>

Spring 2026 Data Lifecycle Training

Plan & Design

March 11 
The Lifecycle of
Scholarly 3D Data

April 8 
Data Management
with DMP Tool

May 6 
Offboarding for
Research Projects

Collect & Analyze

February 25 
Interact with RShiny

March 12 
Intro to MATLAB

March 26 
Public Health GIS

April 15 
Intro to Git & GitHub

April 29 
Prepare and Share Your
Software

April 30 
Tidy Up Your Data!

May 14 
Intro to Python

Store & Evaluate

February 26 
Finalize, Submit, and
Transfer Your Data

March 10 
Introduction to the
General Records Schedule

March 24 
Managing Your Paper
Records: Off-Site Records
Storage

April 21 
Managing Your Electronic
Records: Shared Drives
and Email

Share & Publish

February 5 
Share and Publish
Data with OSF

March 10 
The Making of a Data
Availability Statement

March 18 
Publication Perfect

May 20 
Data Sharing with
Harvard Dataverse

 In-person
 Virtual

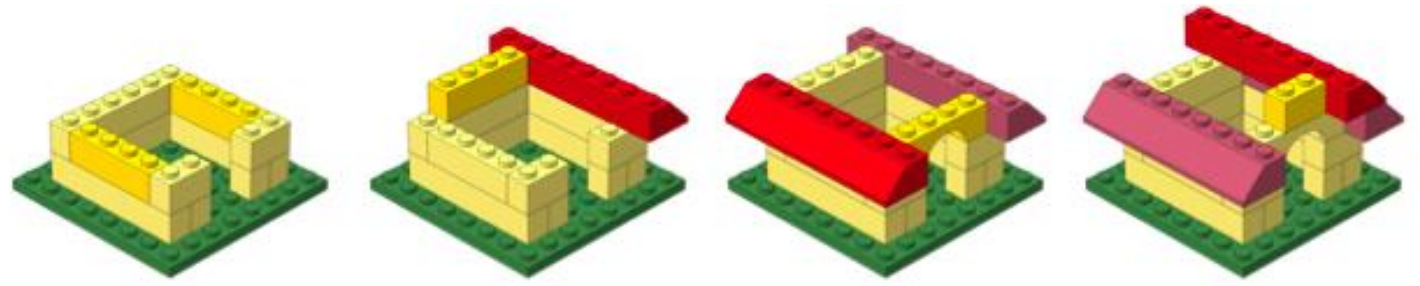


Learn More & Register
bit.ly/rdmwg-calendar



<https://datamanagement.hms.harvard.edu/training-events/rdmwg-calendar>

Keep building!



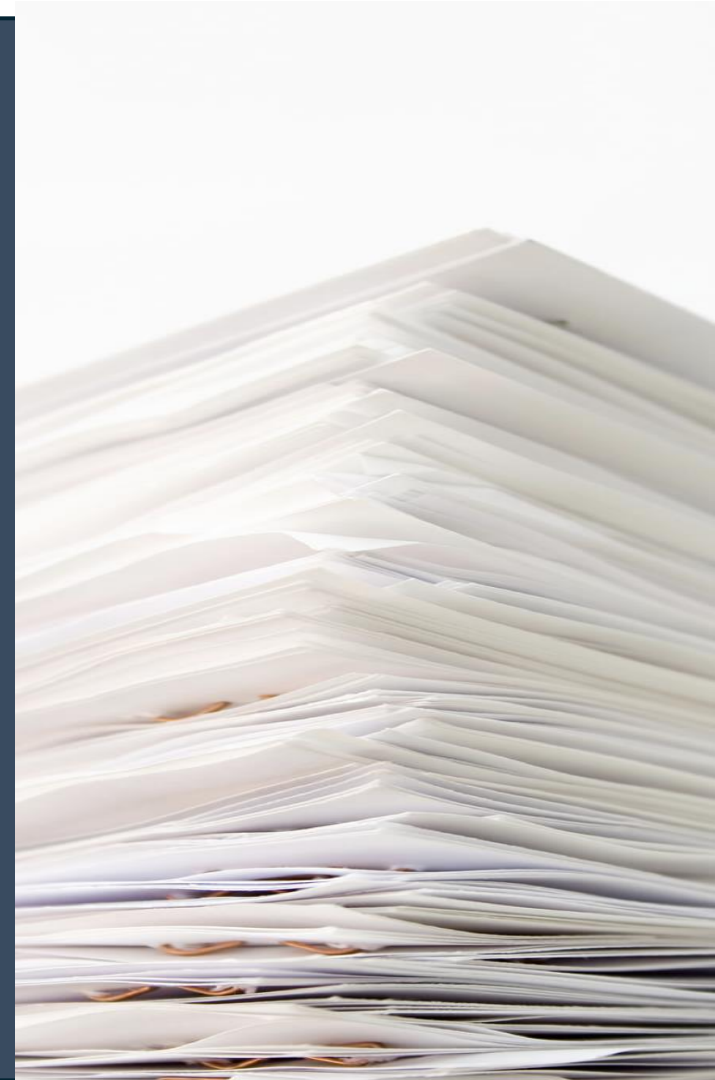
Workshop	Dates	Time	Location	Registration Page*
Introduction to Python	April 14, 17, 21 & 24, 2026	9:30am–12pm	Zoom	TBA
Introduction to Python	July 7, 10, 14 & 17 2026	9:30am–12pm	Zoom	TBA
Introduction to R	August 28, September 1, 4 & 8 2026	10am–12pm	Zoom	TBA
Introduction to Python	November 10, 13, 17 & 20 2026	9:30am–12pm	Zoom	TBA

Workshop	Pre-requisite*	Dates	Time	Location	Registration Page**
Introduction to Spatial Transcriptomics	Introduction to R	May 5, 8, 12 & 15, 2026	9:30am–12pm	Zoom	TBA
Introduction to scRNA-seq (Python-based)	Introduction to Python	August 11, 14, 18 & 21, 2026	9:30am–12pm	Zoom	TBA
Introduction to Spatial Transcriptomics	Introduction to R	October 6, 9, 13 & 16, 2026	9:30am–12pm	Zoom	TBA
Tools for Reproducible Research	Introduction to R	October 27, 30 & November 3, 2026	9:30am–12pm	Zoom	Register here

<https://hsph.harvard.edu/research/bioinformatics/training/training-schedule/>

Talk to us early!

Involvement in study design to optimize experiments



More Information

- ❖ *HBC training materials: <https://hbctraining.github.io/main>*
- ❖ *HBC website: <https://hsph.harvard.edu/research/bioinformatics/>*

Contact Us

Sign up for our mailing list:

<https://tinyurl.com/hbc-training-mailing-list>

- ❖ *HBC training team:* hbctraining@hsph.harvard.edu
- ❖ *HBC consulting:* bioinformatics@hsph.harvard.edu